

Interviews and Interviewers 1:

*Recruitment and Respondent Selection, Interviewing
Techniques*

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Fundamentals of Data Collection I

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Interviews vs. Self-administration

- Interviewers can improve quality
 - Recruit sample members who might not otherwise participate
 - Observe surroundings (in-person interviews) to provide context for responses
 - Motivate *Rs*, clarify *Qs* and response task
 - Collect biomarker (e.g., blood) and physical (e.g., grip strength) data in person
- Interviewers can introduce error when
 - responses elicited by > 1 *Iwer* are differently distributed (creating variance)*
 - their “observable” attributes, e.g., race, gender, age, facial expressions, are relevant to *Q* topic and respondent’s value (creating bias)
- Interviewers are a human resource
 - Expensive: Single greatest budget line in surveys with interviewers
 - Cannot work around the clock

**Need random assignment of *Rs* to *Iwers* to measure ρ (rate of homogeneity) which is a variable specific population parameter for a given design*

Interviewers and Interviews: Outline

■ Obtaining interviews

- Recruiting sample members
- Within household selection
- Proxy respondents

■ Conducting Interviews (Technique)

- Standardization Debate: Wording vs. Meaning

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- Formal vs. Personal Style
 - Rapport between Interviewers and Respondents
 - Increasing Respondent Motivation

■ Interviewer effects

- Variance: Interviewer Behavior
- Bias: Fixed Attributes

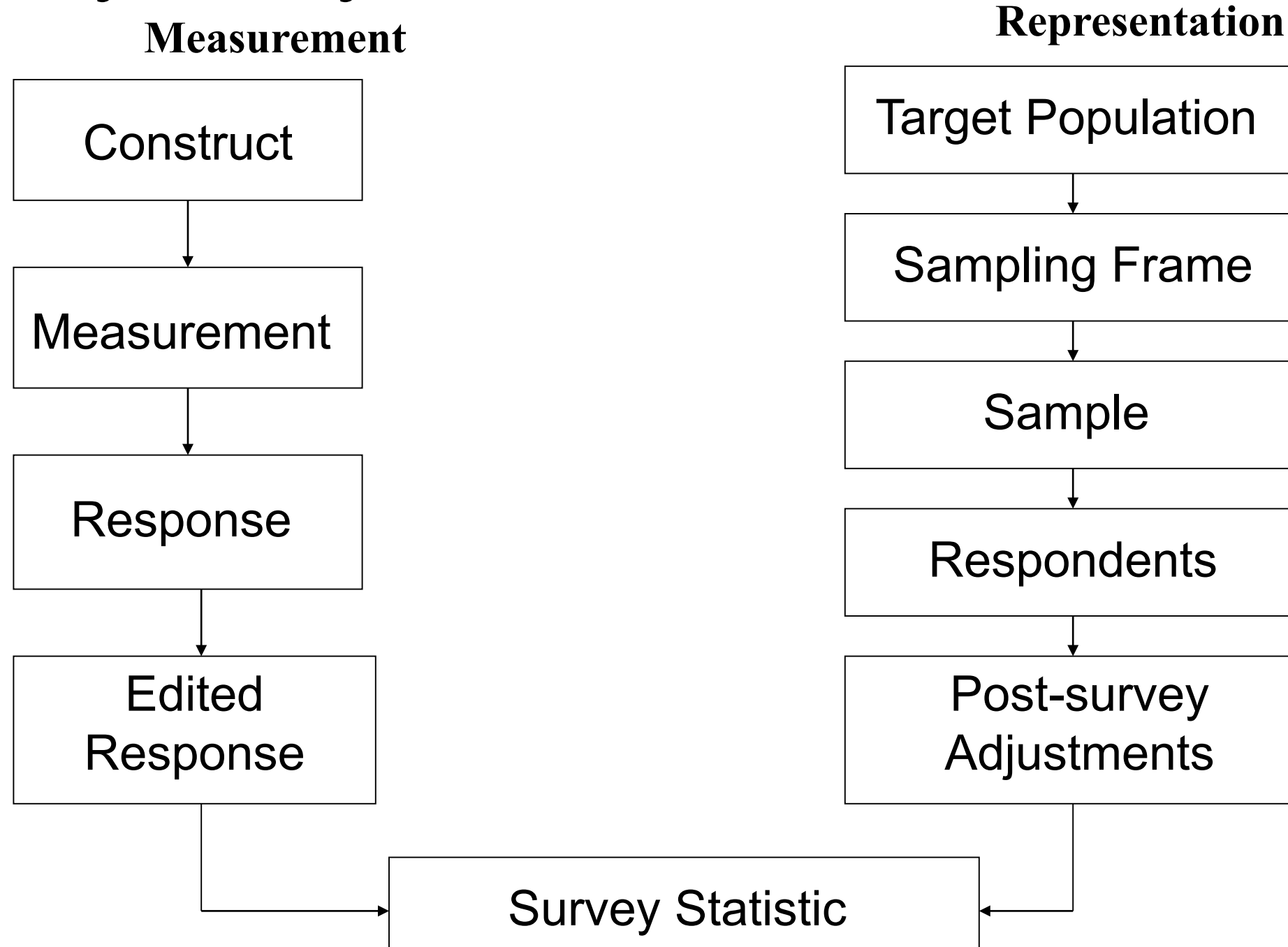
Part I

Part II

Survey Interviewers Play Many Roles

- Each can affect one or more sources of error
- Sampling and coverage
 - Locate and contact sample households
 - Determine eligibility
 - Sample within household: select R
- Recruitment and Participation
 - Invite sample members
 - Gain cooperation, including persuading reluctant R s

The survey life cycle



Survey Interviewers Play Many Roles (2)

- Measurement: Interview selected sample member
 - Ask the survey questions
 - Probe for incomplete answers, clarify questions
 - Record answers
- Interviewers also have an enormous impact on costs and timeliness
 - Usually, the single biggest component of costs
 - Especially if travel is involved (in-person)

Interviewer Effects on Sampling and Coverage

- Listing
 - Recording addresses as walk/drive through neighborhood
 - Determine eligibility of housing unit
 - Identify additional or missed housing units
- Within household sampling
 - Listing all members of the household
 - creates a mini-frame from which R is selected
 - Apply within household selection procedures

Listing, Undercoverage and Coverage Bias

- Eckman & Kreuter (2013)
 - Examined two cycles of listing in NSFG
 - 86% of housing units listed first time were listed second time, i.e., 13% undercoverage
 - undercoverage attributed to multiunit status, rural areas, and map difficulties
 - Coverage bias: Units not covered in second listing (but listed in first) different than units covered in second listing on several attributes: birth control use, pregnancies, education and income
 - Survey data available for undercovered units from first listing make it possible to determine bias
- Authors argue undercoverage from listing should concern data users

Interviewer Introductory Behavior

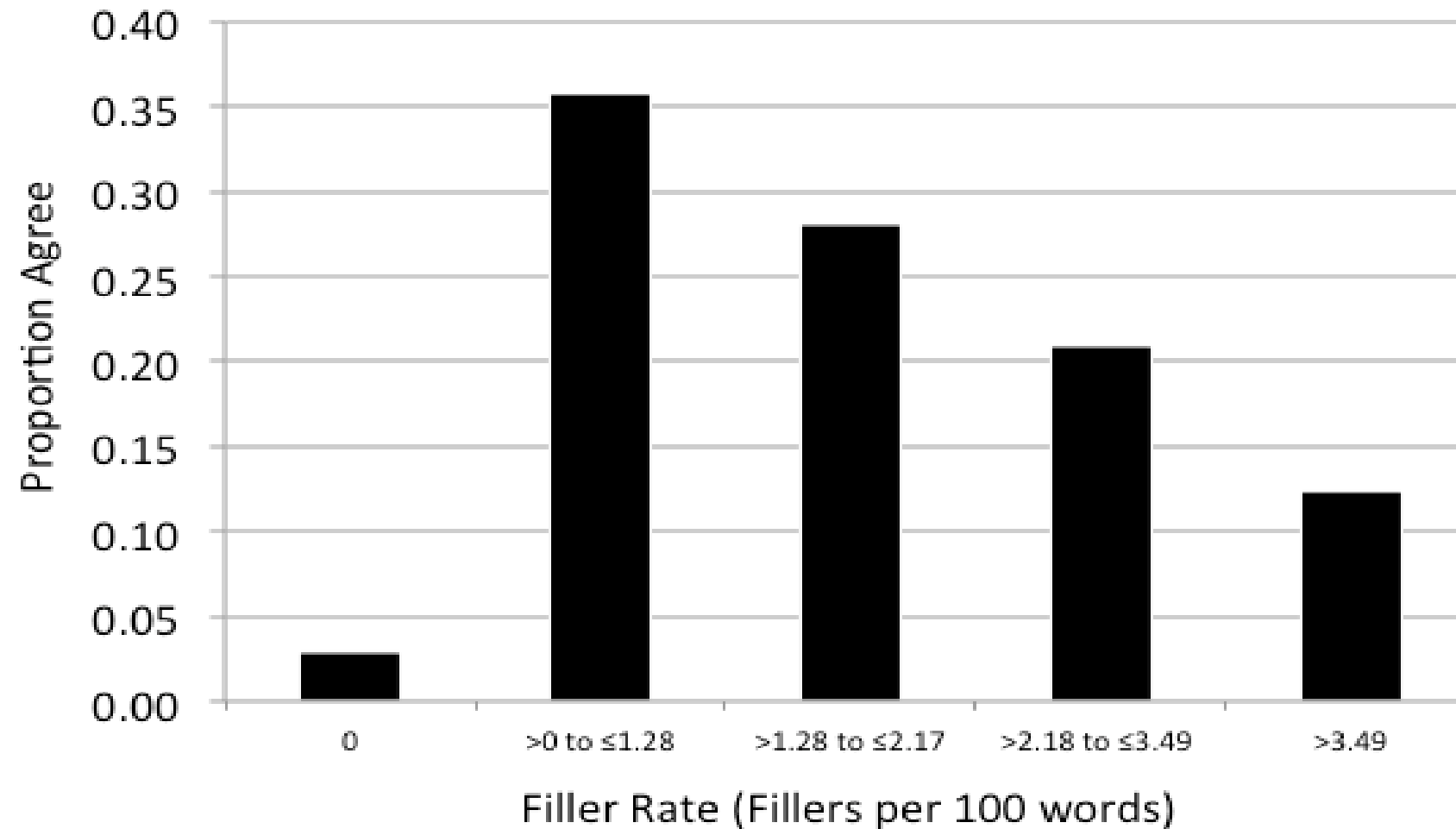
- Tailoring Behavior by Interviewers (Morton-Williams, 1993)
 - Supports the value of lwrs' tailoring their introductions
 - Study design:
 - 14/30 lwrs used a scripted introduction to a face-to-face survey
 - Content of introduction
 - lwr's name
 - Identity of organization
 - Survey description
 - How address was selected
 - Findings:
 - Scripted: 59% response rate
 - Unscripted: 76%

Interviewer Effects on Participation: Speech and Interaction (Example 1)

- *Interviewers* vary in their success obtaining interviews
- Seems likely that how *Interviewers* speak can affect success
- Conrad, et al. (2013)* coded speech behaviors of *Interviewers* and Householders (*HHers*) in 1380 telephone survey introductions
 - Interviewer fillers (e.g. 'um' and 'uh')
 - Overspeech/Interruptions between interviewer and householder
 - Householders' backchannels (e.g. 'uh huh' and 'I see')
- And modeled impact on three outcomes: agree, refuse, scheduled callback

*Conrad, Broome, Benkí, Kreuter, Groves, Vannette & McClain (2013)

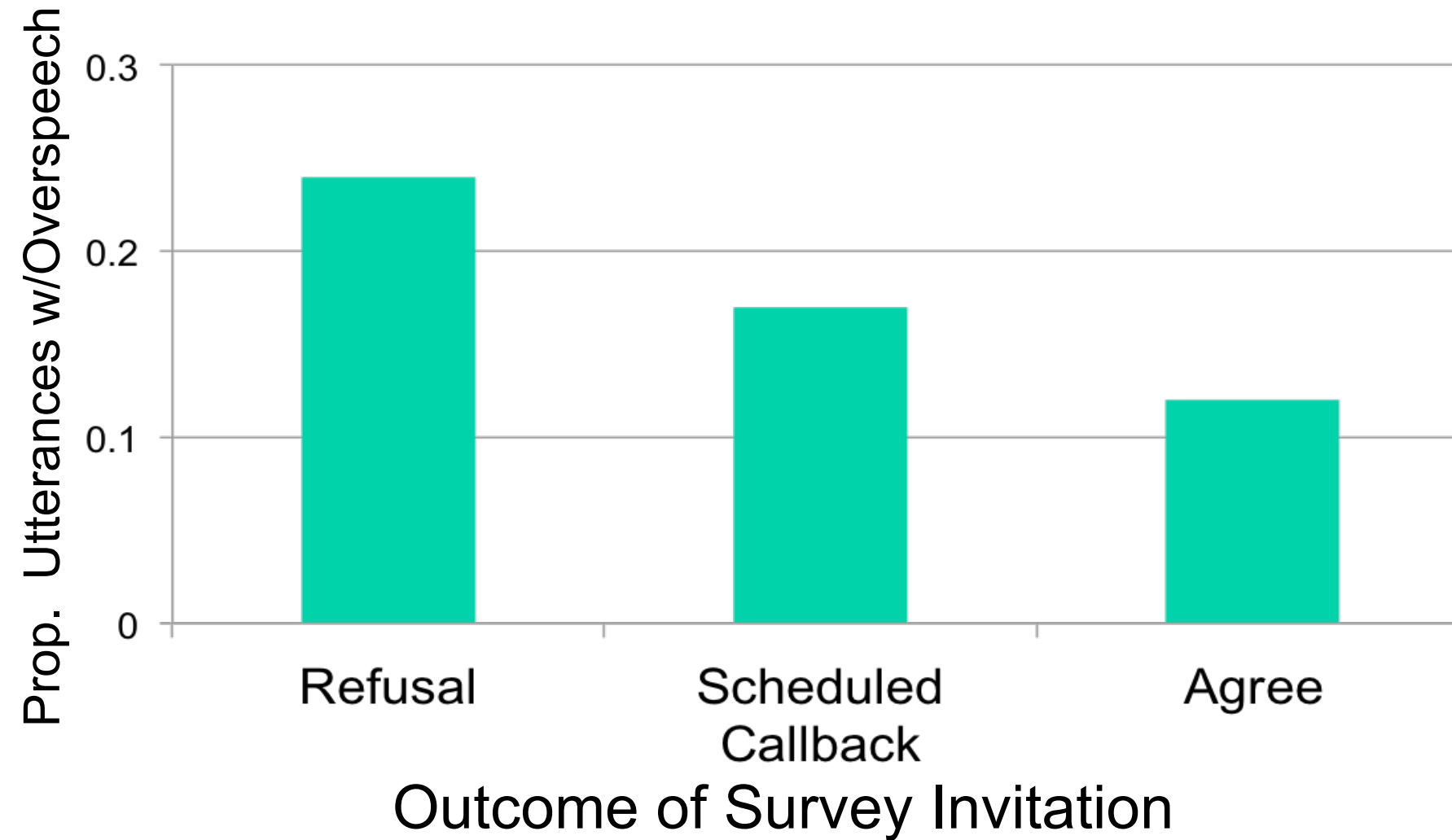
“Agrees” by / Filler Rate*



* / fillers per 100 words

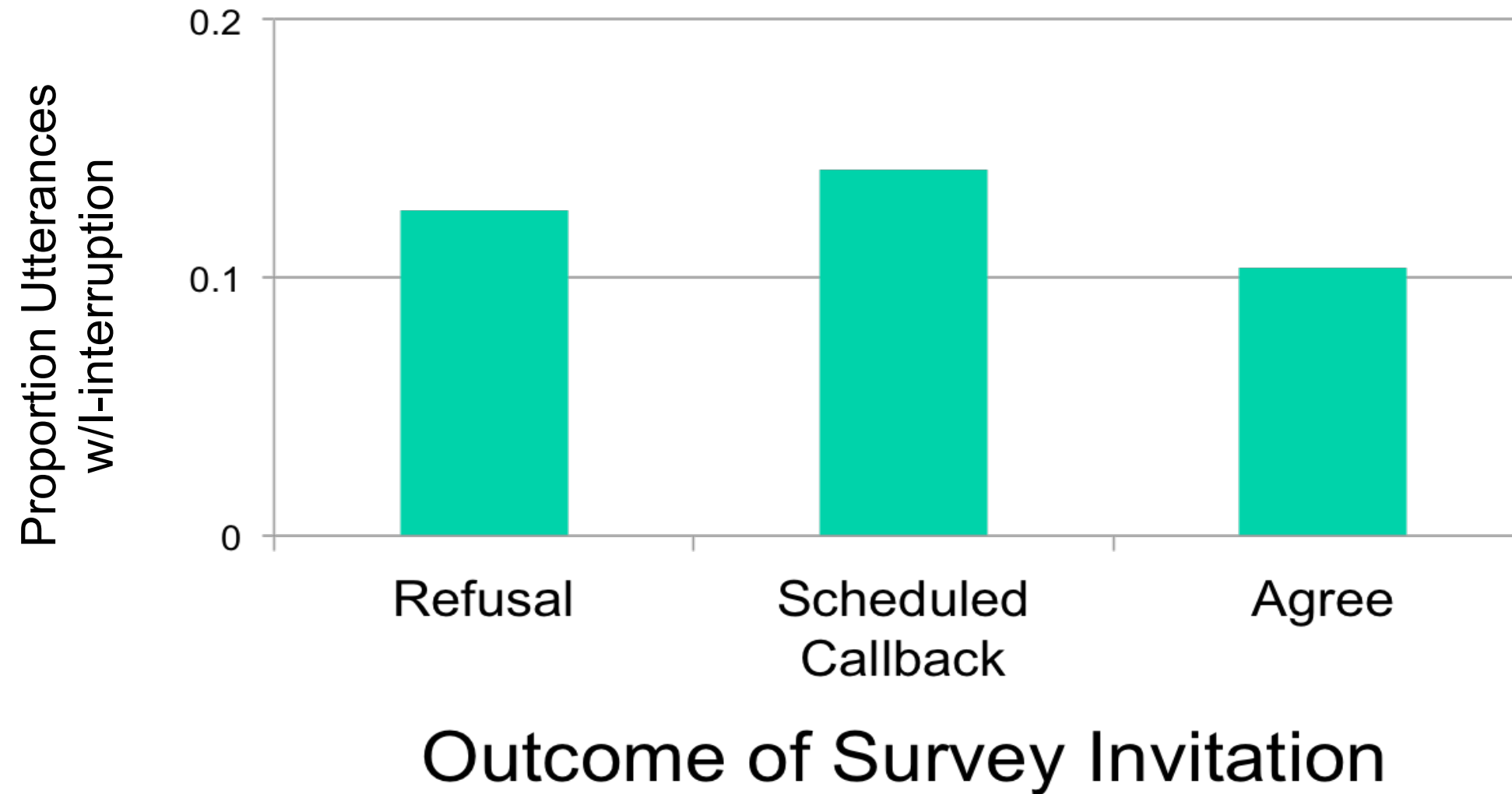
Relation between Overspeech and Participation Outcome

- Most in refusals



Relation between Overspeech and Participation Outcome (2)

- Refusals deferred via interviewer interruptions



Relation between Backchannels and Participation Outcome

- Most dialog is punctuated by utterances such as *uh huh*, '*mmm*', '*yes*', '*I see*' and '*okay*' – known as “backchannels”
- Generally assumed to signal to speaker that listener is engaged and attending to the speaker, implicitly consenting to the speaker's continued speaking
- If prospective *R*'s production of backchannels during survey invitation similarly signals consent to *lwer*'s continued speaking, does this increase likelihood of participation?

Relation between Backchannels and Participation Outcome (2)

- Conrad et al. (2013) found proportion *R*-backchannels differed by outcome: Agree (.14) > Scheduled Call Back (.09) > Refusal (.04)
- If *HHers* volunteer likely willingness (e.g., “All right, let’s do it”) vs. likely refusal (e.g., “I really don’t like to do these things”), how does proportion backchannel change after such “turning points”?

<i>Turning point statement</i>	<i>Position of backchannels</i>		<i>Mean proportion householder backchannels</i>
	<i>Before</i>	<i>After</i>	
Willing	0.183	0.293	0.199
Not interested	0.095	0.045	0.068
Mean proportion of backchannels	0.10	0.13	0.11

Interviewer Effects on Participation: Speech and Interaction (Example 2)

- Schaeffer, Garbarski, Freese, & Maynard (2013)
 - Selected 257 pairs of cases from WLS*, with same propensities to participate (based on data in frame and prior wave participation)
 - For each *declination*, identified the *acceptance* with closest propensity score and paired them
 - Given matched propensity to participate, origins of different recruitment outcomes might be evident in *Iwer-R* interaction
 - Classified both *Iwer* and *R* speech using conversation analytic techniques

*Wisconsin Longitudinal Study

Interviewer Actions in Call Opening Associated with Increased Odds Of Acceptance

Action	Feature of action	Odds ratio	P-value
Greeting token	Hello (vs. hi)	1.44	0.06
Request to speak to sample member	Very polite (vs. not)	1.60	0.07
Order of interviewer identification and request to speak	Interviewer identification first (vs. second)	1.80	0.04

Where did Decisions (Decline, Accept) Occur?

Declination or acceptance occurs...	Outcome	
	Declination	Acceptance
During opening	57	0
During study description	95	4
After request for participation	105	253

Relation between Sample Member's Question-Asking and Acceptance

Feature	Odds ratio	P-value
Sample member asks at least one question	2.07	0.00
Sample member initiates statement or question about the length of the interview	5.44	0.00
Sample member asks at least one wh-type question	0.59	0.02
Sample member asks at least one wh-type question before the request to participate	0.35	0.00
Sample member asks at least one wh-type question after the request to participate	1.64	0.14

Interviewers' Responsiveness to Sample Member Concerns

- Sample members often express concerns about participating
 - timing of call, inability to hear *Iwer* clearly, lack of interest in topic
- Broome (2015) coded interactions in Conrad et al. (2013) corpus for extent to which *Iwer* responded to sample members' concerns
 - Assigned *I*-responsiveness score to each invitation
 - Responsiveness higher in example #1 than example #2
 1. Answerer: "There's a lot of questions that we probably couldn't even answer.
Interviewer: "Well, it's not a test or anything."
 2. Answerer: There's only two of us and my husband's in the shower and I'm in the middle of making dinner.
Interviewer: Well, this is a very important study.
- Responsiveness higher for *Agree* (.63) than *Refuse* (-.32)
 - Highest for Scheduled callback = .80 (surprisingly)

Can /s be trained to more effectively recruit sample members?

- "Interviewers with minimal training can handle the interpersonal aspects of enlisting cooperation and relating to respondents as well as anybody else, in part because we do not know how to train interviewers in these skills very well"

Fowler & Mangione, 1990, p. 118

Training Interviewers to Recruit Respondents

(Groves & McGonagle, 2001)

- Instructed interviewers:
 - To classify typical respondent concerns
 - “I’m too busy right now” => insufficient time
 - To quickly generate appropriate response
 - “This will only take a few minutes”.
 - “When would be a better time?”
- Drilled interviewers until competent
- Interviewers trained in the technique recruited
 - 10% more establishments (study1)
 - 9% more households (study 2)

Respondent Selection

- Sampling unit often a HH because of sampling frame content
 - e.g., street address or landline telephone number
- But analytic focus of survey may involve individuals
- Method needed to randomly select one *HH* to be *R*
- We discuss two *R*-selection methods, but others exist (see Gaziano, 2005)
 - Kish method
 - Birthday methods

Kish (1949) Roster Method

1. List all members of housing unit/household, then check eligibility of members
2. Number the eligible males from oldest to youngest then eligible females from oldest to youngest
3. Use selection table (“cover sheet”) to select respondent
 - Assures all eligible *Rs* in HH have equal chance of selection
 - 8 selection tables randomly allocated to each household

Kish Roster Method (continued)

Selection Table B	
If the number of eligible persons is:	Interview the person numbered:
1	1
2	1
3	1
4 or more	2

Kish Roster Method (continued)

Selection Table D	
If the number of eligible persons is:	Interview the person numbered:
1	1
2	2
3	2
4 or more	3

Kish Selection Tables (6-Table Version)

Table	Proportion of households	If the number of eligible adults is ..., interview adult number ...			
		1	2	3	4+
A	1/4	1	1	1	1
B	1/12	1	1	1	2
C	1/6	1	1	2	2
D	1/6	1	2	2	3
E	1/12	1	2	3	3
F	1/4	1	2	3	4

Kish Roster Method: Pros and Cons

■ Advantage

- Most precise selection method: each HH member has equal (and known) probability of selection
- Can be automated
- Can be extended to oversample groups of interest (e.g., teenagers, males, etc.)

■ Disadvantages

- Requires gender and age of all eligible HH members
- Intrusive, time consuming
- Subject to interviewer error

Birthday Methods

- Last birthday (Oldendick, Bishop and Tuchfarber, 1988):

May I speak to the person in the household who is 18 years of age or older and whose birthday was most recent?

- Next Birthday (Salmon and Nichols, 1983):
 - Interviewer asks to speak to the adult member of the household who has the next birthday
- Birthday methods not probability method – selection probabilities unknown – but introduces quasi-randomness

Birthday Selection Methods: Advantages

■ Advantages

- Relatively easy to administer
- Equal (but unknown) probability of selection within household
- Less intrusive, faster, and lower effort for HH member than Kish method
- Generates sample that ~random

Last and Next Birthday Selection Methods: Disadvantages

- Exact chance of selection is unknown – “quasi-probability method” (Gaziano, 2005) because don’t know how many eligible sample members in HH
- Subject to reporting error; difficult to verify
- Biemer & Lyberg (2003) recommend against use in rigorous, scientific studies
 - But many serious studies use one of the birthday methods

Comparison of Kish and Birthday Methods

- Binson, Canchola & Catania (2000) compared dropouts in Kish and birthday methods in RDD study
- Overall, Kish performed worst (more dropouts) and Next Birthday performed best

% Dropout: $\frac{\text{LB}}{34.1}$ $\frac{\text{NB}}{32.0}$ $\frac{\text{Kish}}{37.0}$

- But much of the effect occurred before *R* could know what type of screener questions would be asked

% Dropout: $\frac{\text{LB}}{16.1}$ $\frac{\text{NB}}{12.2}$ $\frac{\text{Kish}}{16.6}$
Before end

- Authors attribute to interviewers “telegraphing” what will follow

Hybrid R-Selection Method

- Rizzo, Brick & Park (2004) propose method that is minimally invasive, reduces recruitment effort, and usually explicit about $p(\text{selection})$
- Ask R in screener how many adults in household
 - If $N = 1$, screener R selected; end of process
 - If $N > 1$, randomly sample with $p(\text{selection}) = 1/N$
 - If $N = 2$ and screener R not selected, then other selected
 - If $N > 2$ and screener R not selected, then use Last Birthday method (or Kish when birthdays not known) to select from remaining adults
- Rizzo et al. tested in HINTS (RDD survey of where American adults acquire health info)
 - Male and female proportions close to what expect from CPS

Hybrid Method of Respondent Selection (continued)

	% Households with Screener R selected		
Num. Adults in Household	Expected	Actual (HINTS)	% HHs with additional sampling
1	100.0	100.0	0.0
2	50.0	50.6	0.0
3	33.3	34.5	65.5
4	25.0	25.4	74.6
5+	11.8	15.5	84.5
Total	60.9	61.4	12.2

Choosing Among Respondent Selection Methods

- Response errors affect all methods
 - Kish method requires enumeration of HH which may entail measurement error
 - Birthday methods reveal purpose of questioning; easy to manipulate if, e.g., HH member does not want to participate
- Full listings of households require explaining the rationale of *R*-selection process to household member
 - May lower overall response rates
- Different organizational cultures regarding preferences for different methods; different researcher preferences

Proxy Respondents

- Is it okay to collect data about selected R from someone other than selected R , *i.e.*, a proxy R ?
- Several advantages
 - Less time and, therefore expense, to use proxy reports (e.g., CPS reduces cost by about 12%)
 - Using proxy R s results in higher response rates
- General belief that self-reports are better than proxy reports
 - Proxies' *response process* thought to rely more on general knowledge than actual events (Blair, Menon and Bickart, 1991; Sudman et al., 1994) compared to self-reporters

Self/Proxy Differences as Actor/Observer Differences

- Schwarz and Wellens (1997) tested this idea in Self/Proxy context:
 - Self-reporters' answers based on memory for specific autobiographical events
 - Proxy *Rs*' reports draw more on general knowledge (dispositions) of "the other," i.e., the intended/selected *R*
 - Intended/selected *Rs*' answers (which are about themselves) should reflect more *inconsistencies* in behavior – departures from usual routine – than in proxy reports

Schwarz and Wellens (Exp. 3):

- 24 couples asked to report frequency of either own or partner’s typical behavior and highest and lowest weekly frequency during last month
- Reports about self should deviate more from “typical” frequencies than proxy reports
- This is the case: overall, 4.38 vs. 2.87 for self vs. proxy deviations

	Self		Proxy	
	Highest	Lowest	Highest	Lowest
Library	+3.41	-4.32	+2.13	-1.91
Snacks	+7.13	-6.14	+4.67	-4.93
Being Late	+2.61	-2.98	+1.82	-1.61

Mean difference between typical and highest or lowest behavioral frequency

- How data quality is affected by proxy responses may depend on topic of question

Summary: Obtaining Interviews and Selecting Respondents

- Interviewers play several roles; each can affect survey error for better or worse
 - Sampling, coverage, nonresponse and measurement
- Recruitment:
 - Interviewers vary in success; manner of speech and sample member's reaction seems to matter
- Within-household sampling (assuming a household sampling frame)
 - Methods vary in ability to specify $p(\text{selection})$, intrusiveness, efficiency
 - Essential for full implementation of sample design
- Self-proxy issues
 - Response processes differ between self and proxy R s
 - Cost, nonresponse, and measurement error trade-offs

Interviewing Techniques: Outline

1. Standardized Interviewing and the Standardization debate

- - Standardized wording vs. standardized meaning

2. Formal and personal styles of interviewing

- Pros and cons increasing rapport between *I* and *R*

3. Commitment, instructions, and feedback

- Impact on data quality; extension to online self-administration

4. Interviewer effects

Standardization Debate:

Goals and Assumptions of Standardized Interviewing

- Promote *I*wers being interchangeable
 - Should not matter who asks the *Q*s
- Minimize interviewer-related error (variance)
 - Assuming *R*s randomly assigned to *I*wers and standardized administration, distribution of answers should not vary by *I*wer
- Promote comparability of answers
 - if *Q* (stimulus) is identical for all *R*s, different answers from different *R*s cannot be due to *Q* wording
 - differences assumed to be due to actual differences between *R*s

Standardized Interviewing

- *I*wer reads Qs exactly as worded in questionnaire
- If *R*'s utterance is anything other than acceptable response, administer neutral (or non-directive) probe, e.g., “Let me repeat the question”
 - *I*wer chooses whether/how to probe so has some discretion, i.e., not fully standardized
 - Variation in probe wording involves non-substantive info, i.e., Q wording always the same
- Ask Qs in the order they appear in questionnaire
 - Otherwise, context of Q differs changing stimulus; makes answers hard to compare
- Ask every Q even if *R* has previously volunteered relevant info
 - *R* might answer Q differently than what the previously volunteered info implies
- Be professional: avoid reactions that *R* might interpret as evaluative

Examples of Neutral (Non-directive) Probes

- After *R* answers *Q*:
 - “Is that ‘yes’ or ‘no’?”
 - “Could you be a little bit more specific?”
 - “Which comes closest to the way you feel?”
- After *R* requests Clarification:
 - “Let me reread the question”
 - “I’m sorry, but I don’t have that information”
 - “Whatever it means to you.”

Criticism of Standardized Wording

- The chief criticism is that standardized wording prevents *conversational grounding* (e.g., Clark, 1996; Suchman & Jordan, 1990)
 - Grounding: Speaker and listener talk back and forth until they agree they understand each other well enough for purposes of current task
 - In survey interviews, establishing grounding can require different wording for different *R*s, i.e. wording not strictly standardized
- Goal should be to produce uniform interpretation across *R*s so *lower* should aim to standardize meaning rather than wording
 - While standardized wording may promote reliable responses (e.g., same answers from same *R*s on different occasions), not necessarily valid answers
 - assuring that *R*s understand *Q* in the same way – which grounding helps achieve – should promote valid (i.e., accurate) responses.

When is Standardized Wording Sufficient?

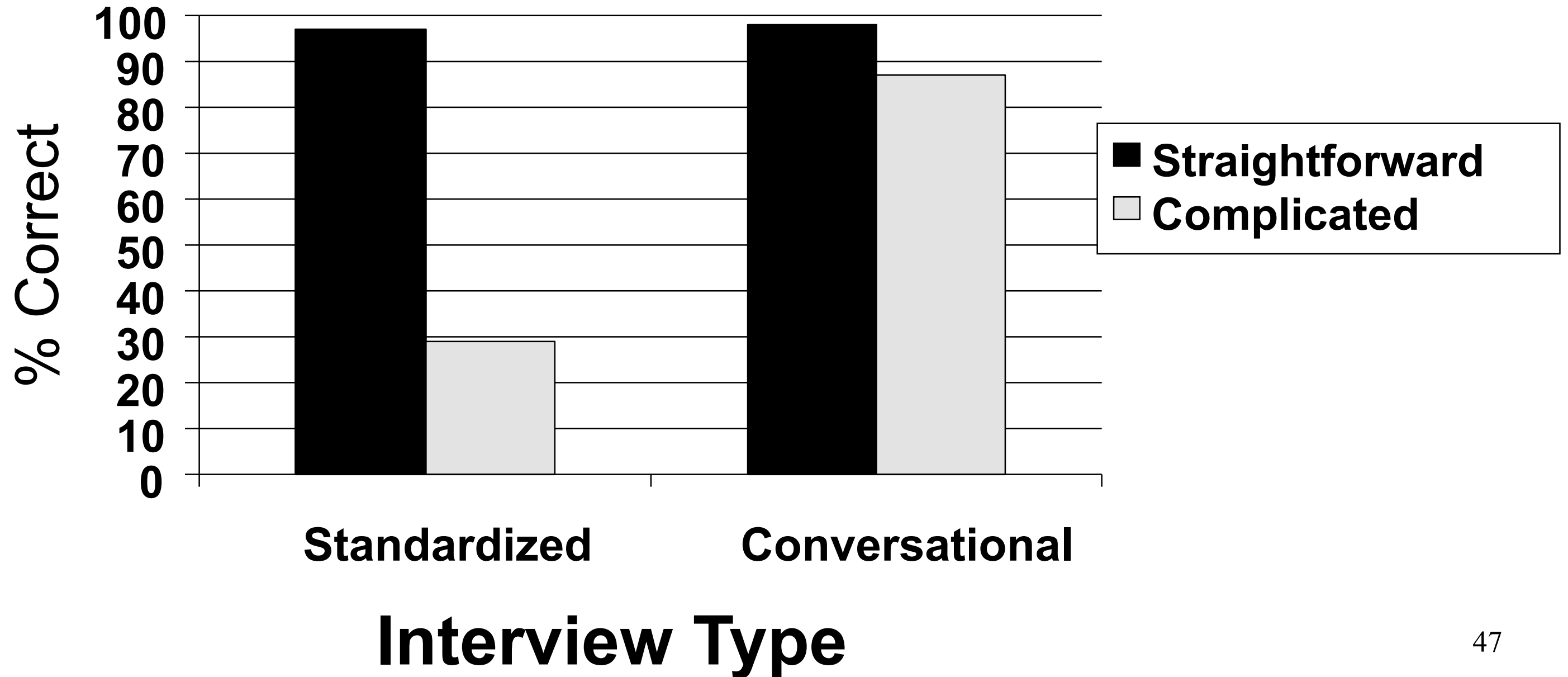
- When all *Rs* interpret question as intended
 - Everyone would probably agree that if a married couple with two young children live in a house, and no one else ever sleeps there, then four people live in the house
- But what about situations in which *Rs* do not know what to include or exclude?
 - What if one child is away at college?
 - What if a nanny sleeps in the house most nights, or a border is temporarily living in the house?

Standardized vs. Conversational Interviews

- Schober & Conrad (1997) asked *Rs* (in lab) 12 Qs from US gov. surveys
- *Rs* answered on basis fictional scenarios that allowed researchers to measure response accuracy
 - ½ scenarios, Q ambiguous w/o a definition: *complicated mappings*
 - e.g., answer Q about furniture purchases on basis receipt for buying floor lamp
 - other ½, Q clear (unambiguous) w/o definition: *straightforward mapping*
 - e.g., receipt for purchase of end table
- Two different interviewing techniques:
 - Standardized Interviewing (SI) where *lwer* cannot deviate from script
 - Conversational Interviewing (CI)*:
 - 1) encouraged *R* to ask questions if did not understand something
 - 2) *lwer* able to clarify word meanings, e.g., whether someone who babysits for multiple families has 1 or >1 job, even if changed wording for different *Rs*

* Called “Flexible Interviewing” in 1997 paper; came to be called “Conversational Interviewing” in later papers

Response Accuracy



CI Duration Greater than SI

- CI can produce greater response accuracy than SI
- but also produces longer interviews

	Standardized	Conversational
Median	3.41	11.47
Min	2.48	6.10
Max	5.99	35.44

Median Interview Duration in Minutes

- May result in fewer cases for fixed cost
- Size of speed-accuracy tradeoff depends on # complicated situations
- CI-SI Δ interview durations swamped by overall recruitment time

How Often Does Conversational Interviewing Help?

- Assessed through difference in response accuracy* between CI and SI in a national sample (Conrad & Schober, 2000)
 - Response quality measured as change between initial standardized and either subsequent standardized or conversational interview
- Logic:
 - *R*'s interpretation in SI (i.e., no clarification available) may be changed (corrected) by clarification in subsequent CI leading *R* change answer in second (conversational) interview
 - If initial interpretation (misconception) persists in subsequent SI, less likely to change answer

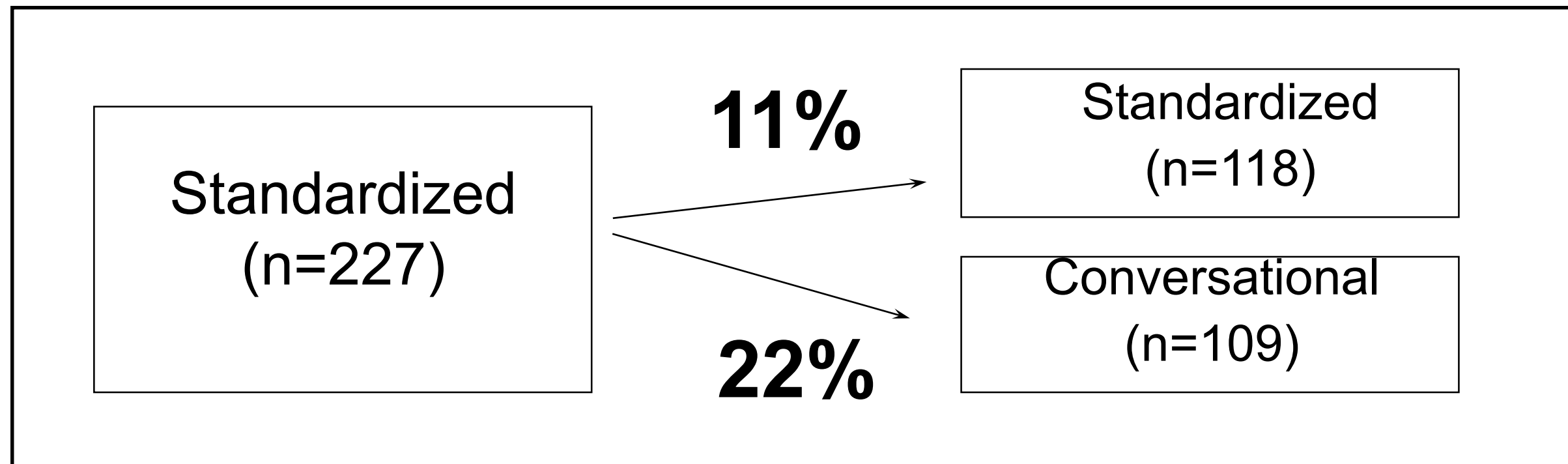
*indirect measure

Results: Response Change

- More responses changed when second interview was conversational than standardized

INTERVIEW 1

INTERVIEW 2



Standardization Debate: Summary

- *Conversational interviews* can substantially improve response accuracy compared to *standardized interviews*
 - when scope of questions is ambiguous, e.g., what should be included and what should be excluded
 - because conversational interviewers can ground the meaning of the terms and phrases in Q
- Grounding takes time so duration of conversational interviews greater than strictly standardized interviews
- Does Standardized Interviewing actually reduce interviewer variance?